

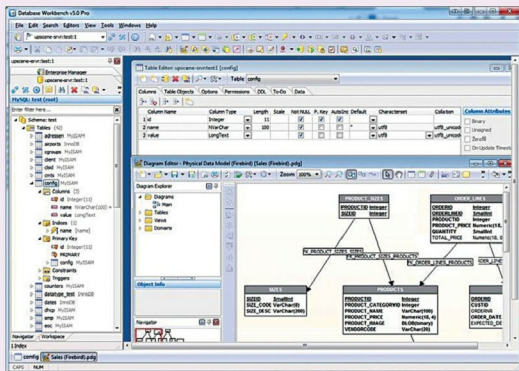


Fig. 3: Database workbench for MariaDB

ently logged in. Below that you will see the database being used. Different existing components of the database like tables, views, functions, events, queries, reports and backups can be viewed.

The middle of the interface is the main workspace, where queries can be coded and results viewed. MariaDB follows simple structured query language (SQL) syntax. On successful execution of a query, results in table format are displayed in a new tab on the workspace. Table Relatedational diagrams and key-based connections can be viewed in the database workbench, providing a clear graphical interface.

Features in Version 10.2

Going by user reviews, MariaDB is one of the smoothest RDBMS. Its read and write functions are quick with fast retrieval of query data. Support for all types of DBMS features like user-defined functions, stored procedures

and views makes this DBMS very useful.

As mentioned earlier, MariaDB comes with multiple storage engines. Version 10.2 uses InnoDB as the default storage engine replacing XtraDB from the earlier iterations.

Apart from that, MariaDB adds TokuDB for fractal index storage, MyISAM for non-transactional data storage, Sequence storage engine, SphinxSE for text search, Spider storage engine, Cassandra and CONNECT for legacy database, and NoSQL backends. This also means MariaDB has a certain amount of NoSQL support. MyRocks is another storage engine that allows big data compression without compromising the speed.

Other than storage engines, there is a massive list of third-party plugin support as well. For instance, Version 10.2 supports AWS key encryption plugin. There are plugins for Archive, Audit,

Client Statistics, Memory, Performance_Schema, Partition, Server Audit and more. To check all the plugins and their integration maturity level in MariaDB, type the following SQL command:

```
SELECT plugin_name, plugin_version, plugin_maturity FROM information_schema.plugins ORDER BY plugin_name;
```

JSON functional-ity has also been added and supports CREATE TABLE command. Support for common table expressions eliminates

the need of temporary tables for complex SQL.

The latest version has added compatibility for multiple triggers on the same event, which can be a handy feature. Moreover, a Flashback feature has been added. It can do a rollback function for databases, tables and instances to an old snapshot. Optimal import and export of CSV and XML files provide added advantage.

The final word

MariaDB developers plan to add rollback on DML statements as well. MariaDB's scalability and versatility have made it the RDBMS of choice for users of every scale. The open source community brings transparency to developmental steps and upgrades made.

Upgrades and security features are available with proper documents and release notes. With a strong community support and global usage, MariaDB is a great database system to go ahead with. **EFY**